Onkar Jadhav

AI / Full-Stack Software Engineer
Pune, Maharashtra
B.Tech Computer Science, PCCOE (9.08 CGPA)

+91-7058604679
onkarj012@gmail.com
github.com/Onkarj012
linkedin.com/in/onkarj012

Summary

Final-year CS student (9.08 CGPA) specialising in LLMs, RAG architectures, and NLP system design with production engineering experience. Built Lore Engine — a full-stack RAG platform with hybrid BM25 + pgvector retrieval over 6,000+ pages, section-aware chunking, streaming chat with inline citations, and a deploy-gated eval harness measuring end-to-end answer accuracy. Shipped Stride (5–10 active users) with a natural-language understanding layer, multi-modal input, and five domain agents over behavioral memory. Built PromptForge for agentic prompt engineering and cross-model benchmarking. Strong in Python, FastAPI, RAG, NLP, vector databases, and LLM integration.

Education

•Bachelor of Technology in Computer Science *Nov 2022 – Jun 2026*
Pimpri Chinchwad College of Engineering, Savitribai Phule Pune University CGPA: 9.08 (up to 7th semester)

– Relevant Coursework: Machine Learning, Data Structures & Algorithms, Database Management Systems, Operating Systems, Computer Networks, Object-Oriented Programming

Experience

•Stride — AI-First Adaptive Wellness Platform *April 2026 – Present*
Founder & Lead Engineer Remote

– Built and shipped Stride to production as sole engineer; running live with **5–10 active trial users** driving iterative product feedback.
– Architected an NLU extraction layer converting multi-modal input (text, voice, image, barcode) into structured data without rigid templates — demonstrating NLP pipeline design across diverse input modalities.
– Built five specialised domain agents sharing a behavioral memory layer, with multi-provider LLM routing via OpenRouter across Anthropic Claude, OpenAI GPT, and Gemini — enabling model-optimised selection per agent type.

Engineering Projects

•Lore Engine — *Full-Stack RAG Platform with Hybrid Retrieval & Eval Harness* *2026 – Present*
Next.js, FastAPI, PostgreSQL, pgvector, BM25, Celery, Redis, Cloudflare R2, OpenAI, Anthropic Claude

– Architected a full-stack RAG application ingesting **6,000+ MediaWiki pages** into a hybrid retrieval pipeline (BM25 + pgvector via reciprocal rank fusion) — specifically chosen over pure semantic search to handle proper-noun disambiguation and exact entity matching that transformer models fail on.
– Designed section-aware chunking at H2/H3 wiki boundaries (not naive fixed token windows) with enriched `[Series]` `[Page]` `[Se` metadata per chunk, improving retrieval precision for fine-grained NLP queries.
– Established a **50-question per-series ground-truth eval harness** measuring retrieval MRR and end-to-end answer accuracy, gating every deploy; validated against the Harry Potter wiki (∼6,000 pages).
– Shipped two AI surfaces on a shared retriever: a streaming chat interface with inline citations and a hierarchically generated reference book — demonstrating conversational AI pipeline design end to end.

•PromptForge — *Agentic Prompt Engineering & Cross-Model Benchmarking Platform* *Jan 2026 – Feb 2026*
Node.js, Express.js, BullMQ, Convex, OpenRouter, React.js

– Architected a creator–critic–evaluator agentic pipeline for iterative prompt improvement, scoring clarity, completeness, constraint coverage, and token efficiency across versioned revision trails.
– Implemented cross-model routing across OpenAI, Anthropic, Gemini, and open-source models via OpenRouter for objective quality, latency, and cost-per-output comparisons for data-driven model selection.
– Built a fault-tolerant async backend (BullMQ workers, idempotent retries, Convex persistence) sustaining reliable prompt regression testing without job loss on LLM provider timeouts.

Technical Skills

AI / ML: RAG Pipelines, LLM Integration, Prompt Engineering, NLP, Embeddings, Semantic Search, pgvector, BM25, Reciprocal Rank Fusion, Agentic Workflows, Behavioral Memory, LLM Evaluation, TensorFlow, OpenRouter
LLM Providers: Anthropic Claude, OpenAI GPT, Gemini, Groq, Deepgram Nova-2, whisper.cpp
Languages: Python, TypeScript, JavaScript, SQL, C++, Java
Backend: FastAPI, Node.js, Express.js, BullMQ, Celery, Redis, REST APIs, WebSockets
Frontend: React.js, Next.js, Tailwind CSS, HTML, CSS
Databases: PostgreSQL (+ pgvector), MongoDB, Convex
Cloud & DevOps: AWS, Cloudflare R2, Git, GitHub, async job pipelines
System Design: Hybrid retrieval, multi-agent coordination, event-driven architecture, conversational AI pipelines